Raw Materials & Composition

- **Bond Paper (Grade A & TMP)**
  - Made from **virgin wood fibres** (hardwood + softwood) for Grade A.
  - TMP (Thermo-Mechanical Pulp) uses **steam + mechanical grinding**, retaining lignin for higher yield and lower cost.
  - Typically **uncoated**, smooth texture, durable.
- **Coated Paper**
  - Base paper coated with **pigments** (calcium carbonate, clay, talc).
  - **Binders** (starch, latex) hold pigments.
  - Additives (resins, polyethylene) for water resistance.
  - Produces glossy, matte, or silk finishes

Manufacturing Processes

- **Bond Paper**
  - **Chemical pulping (kraft process)** for Grade A → removes lignin, strong fibres.
  - **Mechanical pulping (TMP)** for lower-cost paper → retains lignin, bulkier texture.
  - Steps: **Pulping → Formation → Pressing/Drying → Calendaring → Winding → Quality Control**.
- **Coated Paper**
  - **Coating kitchen** prepares pigment/binder mix.
  - **Blade coating** applies uniform layer.
  - **Drying → Calendering → Finishing (gloss/matte/silk) → Cutting/Winding**

Advantages

- **Bond Paper**
  - Environmentally friendly if recycled or FSC-certified.
  - Strong, resistant to ink bleed.
  - Smooth surface → crisp prints.
  - Versatile: works with inkjet & laser printers.
- **Coated Paper**
  - **Enhanced print quality**: sharper, vivid colors.
  - Higher brightness & opacity.
  - Less ink usage due to ink holdout.
  - Resistant to dirt/moisture (depending on finish)

Disadvantages

- **Bond Paper**
  - Not durable without coating (scuffs, tears).
  - Higher ongoing costs (ink/ribbon).
  - Slower printing speeds, especially with impact printers.
  - Heavy bond paper unsuitable for books【uploaded doc】.
- **Coated Paper**
  - Poor writeability (pens/pencils smudge).
  - Gloss glare reduces readability.
  - Higher cost due to coating process.
  - Environmental concerns: plastic-coated papers contaminate recycling.
  - Limited folding/embossing; cracks under stress

Common Uses

- **Bond Paper**
  - Office supplies: letterheads, reports.
  - Legal documents: contracts, forms.
  - Creative projects: scrapbooking, crafts.
  - Education: worksheets, exams【uploaded doc】.
- **Coated Paper**
  - Magazines, journals, brochures, flyers.
  - Catalogs, photographs, book covers.
  - Packaging (moisture/chemical resistance).
  - Marketing collateral: inserts, branding materials

Key Comparison Table

Feature	Bond Paper (Uncoated)	Coated Paper
Surface	Smooth, uncoated	Gloss, matte, silk finishes
Print Quality	Clear, crisp text	Vivid, sharp images
Durability	Strong, but scuffs easily	More resistant (depending on finish)
Writeability	Excellent	Poor (smudges)
Cost	Lower to moderate	Higher
Eco-Friendliness	Good (recycled options)	Variable (plastic coatings problematic)
Best For	Documents, forms, education	Magazines, marketing, photos

Bond paper is best for **professional documents and everyday printing**, while coated paper excels in **high-quality image reproduction and marketing materials**. Your choice depends on whether durability, cost, or visual impact is the priority.